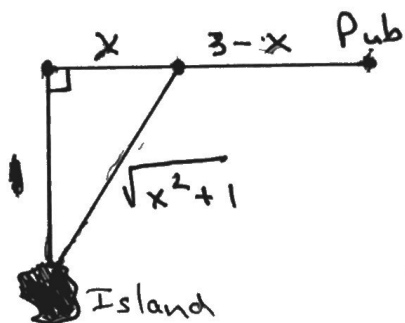


Best Time / Shortest Distance Problems

(1)

Example 1: A man lives on an island 1 km from the shoreline. His favourite pub is 3 km along the shoreline from the point on the shore closest to the island. The man can paddle his canoe at 3 km/hr and can jog 5 km/hr. Determine where he should land in order to reach the pub in the shortest possible time.

Solution:



Let x be the distance from the point on the shoreline closest to the island.

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\text{Total Time} = T_{\text{paddle}} + T_{\text{jog}}$$

$$\text{Total Time} = \frac{\sqrt{x^2 + 1}}{3} + \frac{3-x}{5}$$

$$\left(\frac{\text{Total}}{\text{Time}} \right)' = \frac{1}{3} \cdot \frac{1}{2} \cdot (x^2+1)^{-\frac{1}{2}} \cdot (2x) - \frac{1}{5}$$

$$\frac{2x}{6\sqrt{x^2+1}} - \frac{1}{5} = 0$$

$$\frac{x}{3\sqrt{x^2+1}} = \frac{1}{5}$$

$$\frac{x}{\sqrt{x^2+1}} = \frac{3}{5}$$

$$\frac{x^2}{x^2+1} = \frac{9}{25}$$

$$25x^2 = 9x^2 + 9$$

$$16x^2 = 9$$

$$x^2 = \frac{9}{16}$$

$$x = \pm \frac{3}{4} \quad (\text{reject } -\frac{3}{4})$$

$\therefore$ He should paddle to a point $\frac{3}{4}$ km from the point closest to the island,

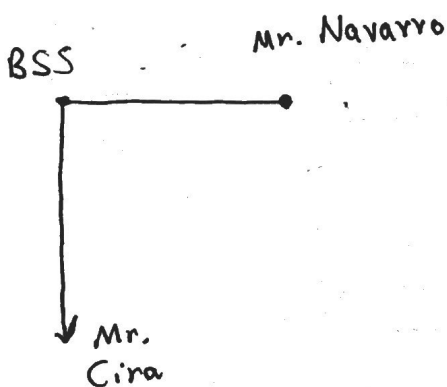
(OR)

$\therefore$ He should paddle to a point $\frac{9}{4}$ km from the pub.

Example 2: Mr. Cira leaves Bayview S.S. at 1:00 pm and drives due south at 80 km/hr. Mr. Navarro is heading due west at 90 km/hr and arrives at Bayview S.S. at 2:00 pm. Determine when they were closest together.

Solution:

Distance = Speed $\times$ Time.



* Mr. Cira's Distance from Bayview S.S. = $80t$

* Mr. Navarro's Distance from Bayview S.S. = $90(1-t)$

$$\therefore d = \sqrt{(80t)^2 + [90(1-t)]^2} = 90 - 90t$$

$$d' = \frac{12,800t - 16,200 + 16,200t}{2\sqrt{(80t)^2 + [90(1-t)]^2}}$$

$$d' = 0$$

$$\therefore 29,000t - 16,200 = 0$$

$$t = 0.559 \text{ hrs } (\text{OR}) \approx 1:33 \text{ pm.}$$